

A review of methionine dependency and the role of methionine restriction in cancer growth control and life-span extension.

Methionine is an essential amino acid with many key roles in mammalian metabolism such as protein synthesis, methylation of DNA and polyamine synthesis. Restriction of methionine may be an important strategy in cancer growth control particularly in cancers that exhibit dependence on methionine for survival and proliferation. Methionine dependence in cancer may be due to one or a combination of deletions, polymorphisms or alterations in expression of genes in the methionine de novo and salvage pathways. Cancer cells with these defects are unable to regenerate methionine via these pathways. Defects in the metabolism of folate may also contribute to the methionine dependence phenotype in cancer. Selective killing of methionine dependent cancer cells in co-culture with normal cells has been demonstrated using culture media deficient in methionine. Several animal studies utilizing a methionine restricted diet have reported inhibition of cancer growth and extension of a healthy life-span. In humans, vegan diets, which can be low in methionine, may prove to be a useful nutritional strategy in cancer growth control. The development of methioninase which depletes circulating levels of methionine may be another useful strategy in limiting cancer growth. The application of nutritional methionine restriction and methioninase in combination with chemotherapeutic regimens is the current focus of clinical studies. Copyright © 2012 Elsevier Ltd. All rights reserved.